

MIDTERM EXAM

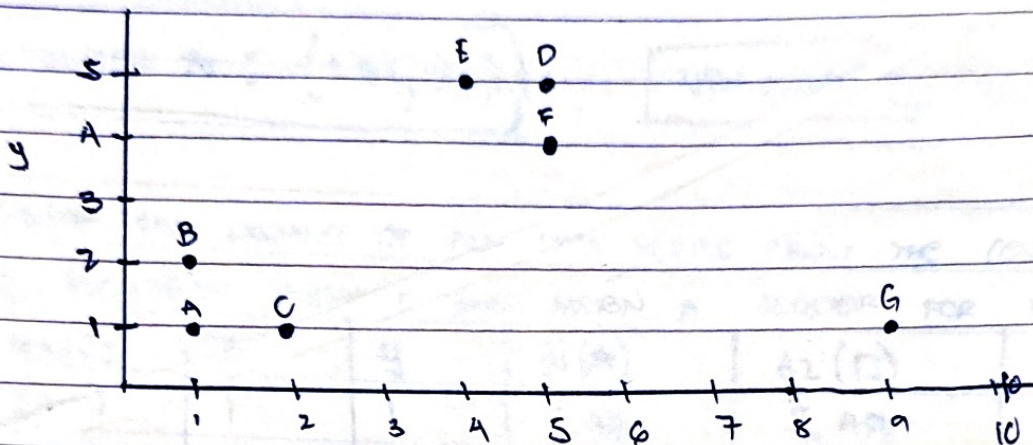
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COM 232

A.

1) PLOT THE DATA IN A SCATTER PLOT.



2)	$(1,1)$ A	$(1,2)$ B	$(2,1)$ C	$(5,5)$ D	$(4,5)$ E	$(5,4)$ F	$(9,1)$ G
A (1,1)	0	1	1	5.66	5	5	8
B (1,2)	1	0	1.41	5	4.24	4.47	8.06
C (2,1)	1	1.41	0	5	4.47	4.24	7
D (5,5)	5.66	5	5	0	1	1	5.66
E (4,5)	5	4.24	4.47	1	0	1.41	6.40
F (5,4)	5	4.47	4.24	1	1.41	0	5
G (9,1)	8	8.06	7	5.66	6.40	5	0

B. KMEANS CLUSTERING : PARAMETERS : $K=2$, CLUSTER : A & D

1)

DATA POINTS	x	y	$d_1(A)$	$d_2(D)$	CLUSTER
A	1	1	0	5.66	★
B	1	2	1	5	★
C	2	1	1	5	★
D	5	5	5.66	0	□
E	4	5	5	1	□
F	5	4	5	1	□
G	9	1	8	5.66	□

B.

2) CALCULATE THE CENTROID/MEAN OF EACH CLUSTER.

$$x_{\star} = \frac{1 + 1 + 2}{3} = 1.33$$

$$x_{\square} = \frac{5 + 4 + 5 + 9}{4} = 5.75$$

$$y_{\star} = \frac{1 + 2 + 1}{3} = 1.33$$

$$y_{\square} = \frac{5 + 5 + 4 + 1}{4} = 3.75$$

NEW CLUSTER $\star$: (1.33, 1.33)

NEW CLUSTER $\square$: (5.75, 3.75)

3) RECALCULATE THE DISTANCE OF ALL DATA POINTS FROM THE CENTROID/MEAN USING EUCLIDEAN DISTANCE AND ASSIGN A CLUSTER FOR EACH DATA POINT

DATA POINTS	x	y	d1($\star$)	d2($\square$)	CLUSTER
A (1,1)	1	1	0.47	5.49	$\star$
B (1,2)	1	2	0.75	5.06	$\star$
C (2,1)	2	1	0.75	5.06 4.05	$\star$
D (5,5)	5	5	5.19	1.46	$\square$
E (4,5)	4	5	4.59	2.15	$\square$
F (5,4)	5	4	4.59	0.79	$\square$
G (9,1)	9	1	7.68	4.26	$\square$

C. HIERARCHICAL CLUSTERING

1) USING SINGLE LINKAGE, DETERMINE THE MERGING ORDER STEP BY STEP AND RECORD THE DISTANCE ON WHICH EACH MERGE OCCURS.

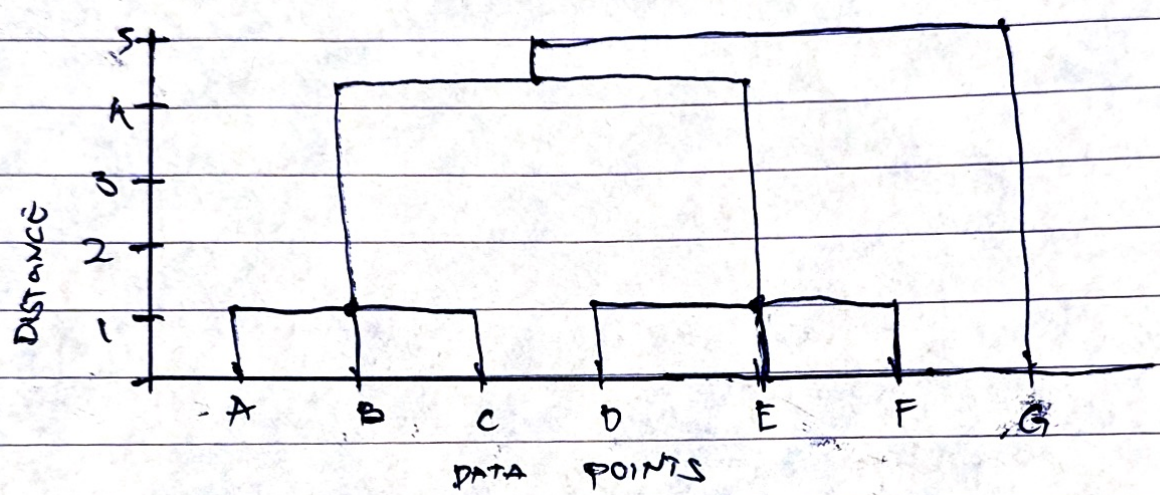
ORDER:

AB: 1	CF: 4.24	AD: 5.66
AC: 1	BF: 4.47	DB: 5.66
DE: 1	CE: 4.47	EG: 6.40
DF: 1	BD: 5	CG: 7
BC: 1.41	CD: 5	GA: 8
EF: 1.41	AE: 5	GB: 5.06
BE: 4.24	FG: 5	

C.7

MERGE ORDER	CLUSTER 1	CLUSTER 2	DISTANCE
1ST	A	B	1
2ND	AB	C	1
3RD	D	E	1
4TH	DE	F	1
5TH	ABC	DEF	4.24
6TH	ABCDEF	G	5

2) DRAW THE CORRESPONDING DENDROGRAM.



D. DBSCAN, PARAMETERS : $\epsilon = 1$, min - samples = 2

DATA POINT	NEIGHBORS	NUMBER OF NEIGHBORS	POINT TYPE	CLUSTER
A	B, C	2	CORE	★
B	A	1	NON-CORE	★
C	A	1	NON-CORE	★
D	E, F	2	CORE	□
E	D	1	NON-CORE	□
F	D	1	NON-CORE	□
G	NONE	0	NOISE	OUTLIER